IT System Health & Availability Monitoring Documentation

1. Overview

The IT System Health & Availability Monitor is a lightweight Python automation tool developed for the **Nourish & Share Food Redistribution Network**.

The purpose of the tool is to provide a simple operational check of the platform's publicly accessible production website.

The monitor checks:

Production URL:
`https://beacon-food-network.web.app`

The tool sends an HTTP request to the production website, measures how quickly the site responds, determines whether the site is online or offline, and generates a basic health report.

This implementation is intentionally limited to **Version 1** and is designed as a standalone IT-support automation component.

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2. Purpose

The purpose of the automation is to help IT Support quickly determine whether the Nourish & Share production website is reachable.

The monitor provides basic availability information without requiring access to:

- User accounts
- Administrator accounts
- Firebase Authentication credentials
- Firestore data
- Private application features
- Production secrets
- Backend credentials

This makes the monitoring process simple and safe for an initial operational availability check.

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3. Scope

The Version 1 monitor performs the following tasks:

1. Sends an HTTP `GET` request to the production website.
2. Checks the HTTP response.
3. Measures the website response time.
4. Records the date and time of the check.
5. Determines the website status.
6. Displays the result in the terminal.
7. Generates a text-based health report.
8. Handles common network and connection errors.

Status determination

The website is considered:

ONLINE

when the server returns a successful or redirect HTTP response in the `2xx` or `3xx` range.

OFFLINE

when the request fails, times out, encounters a network error, or receives an unsuccessful HTTP response.

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4. Project Structure

The automation is maintained as a separate component from the main Nourish & Share application.

```
it-automation/
├── health_monitor.py
├── health_report.txt
└── requirements.txt
```

`health_monitor.py`

Contains the Python monitoring logic.

The script performs the website availability check, measures response time, displays the result, and generates the health report.

`requirements.txt`

Contains the Python dependency required by the monitoring script:

requests>=2.25.0

`health_report.txt`

A report generated by the monitoring script after a health check.

It records the result of the most recent check, including the date, time, target URL, status, HTTP status code, and response time.

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5. Technology Used

The monitoring tool uses:

Technology	Purpose
Python 3	Monitoring automation
Requests	Sending HTTP requests
HTTP/HTTPS	Communication with the production website
Text file	Storing the generated health report

No database connection or Firebase SDK is required for Version 1.

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6. How the Monitor Works

The monitoring process follows this sequence:

Start
↓
Load target production URL
↓

Record current date and time
↓
Send HTTP GET request
↓
Measure response time
↓
Receive response
↓
Check HTTP status
↓
Determine ONLINE / OFFLINE
↓
Display result in terminal
↓
Generate health_report.txt
↓
End

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7. Error Handling

The monitor is designed to handle common failures without terminating unexpectedly.

The following conditions are handled:

Connection Error

Occurs when the system cannot establish a connection to the production website.

Result:

Status: OFFLINE

Timeout Error

The monitor uses a **10-second request timeout**.

If the website does not respond within the configured period, the monitor reports the service as offline.

Network/HTTP Error

Other request-related errors are captured and displayed as part of the health-check result.

Unexpected Error

Unexpected runtime errors are also handled so that the tool can provide an error result instead of failing silently.

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8. Health Check Output

When the production website is available, the monitor produces output similar to:

```
============================================================
            IT SYSTEM HEALTH MONITOR - RESULT
============================================================
Date:          YYYY-MM-DD
Time:          HH:MM:SS
URL:           https://beacon-food-network.web.app
Status:        ONLINE
Status Code:   200
Response Time: XX.XX ms
============================================================
```

[INFO] Health report successfully generated: health_report.txt

The exact response time will vary depending on network conditions and server response time.

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9. Generated Health Report

The monitor generates:

health_report.txt

The report contains:

```
============================================================
                IT SYSTEM HEALTH REPORT
============================================================
Generated On:  YYYY-MM-DD at HH:MM:SS
Target URL:    https://beacon-food-network.web.app
Site Status:   ONLINE
Status Code:   200
Response Time: XX.XX ms
```

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If an error occurs, the report also records the relevant error information.

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10. Installation

Python 3 should be installed on the system.

From the `it-automation` directory, create a virtual environment:

python3 -m venv .venv

Activate it:

source .venv/bin/activate

Install the required dependency:

pip install -r requirements.txt

The virtual environment is used to keep the Python dependency isolated from the system Python installation.

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11. Running the Monitor

After activating the virtual environment, run:

python health_monitor.py

The script will immediately perform one health check against:

https://beacon-food-network.web.app

It will display the result in the terminal and generate or update:

health_report.txt

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12. Testing

The Version 1 monitor was tested against the live Nourish & Share production website.

The successful test confirmed that:

- The production website could be reached.
- An HTTP response was received.
- The website was reported as `ONLINE`.
- The HTTP status code was captured.
- Response time was measured.
- The date and time were recorded.
- A health report was generated successfully.

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13. Security Considerations

The monitoring tool does not require or store:

- Passwords
- Authentication tokens
- API keys
- Firebase credentials
- Firestore credentials
- Administrator credentials
- Donor or recipient credentials

The tool performs only a public HTTP availability check against the production website.

No sensitive credentials should be added to the script or its configuration.

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14. Operational Use

The tool provides IT Support with a basic first-level availability check.

When a user reports that the Nourish & Share website is unavailable, IT Support can run the health monitor to determine whether the public production website is responding.

If the monitor reports:

Status: ONLINE

the website is reachable from the environment where the monitor is running, and the reported user issue may require further investigation.

If the monitor reports:

Status: OFFLINE

IT Support can record the result and escalate the issue to the responsible technical team for further investigation.

The monitor is therefore intended as an **availability-checking tool**, not as a replacement for application-level diagnostics.

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15. Limitations

Version 1 intentionally provides a basic availability check.

It does **not** monitor:

- Firebase Authentication
- Firestore database health
- Cloud Functions
- Private donor functionality
- Private recipient functionality
- Administrator functionality
- Private chat
- Cloudinary
- Email notification services
- Internal application logic

These areas are outside the scope of the Version 1 implementation.

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16. Conclusion

The IT System Health & Availability Monitor provides a simple and reliable first-level operational check for the Nourish & Share production website.

It allows IT Support to quickly determine whether the public website is reachable, measure its response time, identify basic connectivity failures, and produce a timestamped health report.

The implementation is intentionally lightweight and independent of the main application codebase, allowing it to perform its monitoring function without modifying the Nourish & Share frontend, backend, Firebase configuration, authentication system, or database.

Version 1 is considered complete when the monitor successfully performs the production website health check and generates the corresponding health report.